2.2 Optical Drives

2.2.1 Blu-ray Disc

Blu-ray Disc (BD) is a next-generation optical storage medium developed by Sony and others. It uses a blue-violet laser with 405 nanometre wavelength, and offers 25 GB (single layer) 50 GB (dual layer) storage capacity. Blu-ray discs appear as BD-ROM (read-only, pre-manufactured), BD-R (recordable once), and BD-RE (recordable and erasable). The 1X data transfer speed for a Blu-ray drive is 4.5 Megabytes per second. The Blu-ray Disc format is not compatible with the other next-generation optical storage medium, HD-DVD.

2.2.2 Bootable Disc

Optical discs that can be used to start a system are termed bootable. Technically, these discs follow the El Torito extension of the ISO 9660 file format; any motherboard that is compatible with this standard recognises optical drives as a bootable device, and can pass on control to the optical drive to boot the system. Today, bootable optical disks are the norm; you won't find bootable floppy disks except in a few rare cases.

2.2.3 Buffer

Buffer refers to the internal memory present in the drive where data is stored before being written to disc. A 2 MB buffer is common in all drives.

2.2.4 Buffer Underrun Protection

If the buffer becomes depleted midway during a writing session, the media gets corrupted and unusable. Buffer Underrun Protection refers to the techniques used by optical writing devices to prevent the corruption. The writing is temporarily stopped when the buffer is depleted, and resumed when it is filled again. The gap that results is small enough to be tolerated by CD readers, thus ensuring that the CD is not corrupted. Different manufacturers use different names to refer to this technology—like Powerburn (Sony), Super Link (LG), etc.